

MODULE 9 — Billing, Usage & License Enforcement

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module is the **monetization and financial control layer** of the platform.
Its job is brutally simple and non-negotiable:

- Track every billable unit
- Convert usage into money
- Prevent revenue leakage
- Enforce limits without ambiguity

AI calls cost real money per second.
This module ensures **every second is accounted for**.

2. Scope

Applies to:

- Usage metering
- Per-minute billing
- License validity enforcement
- Invoicing
- Payment gateway integration
- Overuse prevention

Does **not** include:

- Pricing strategy decisions
 - Discounts & coupons (optional future)
 - Revenue forecasting
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3. Billing Model (Confirmed)

3.1 Pricing Basis

- Primary billable metric: **Credits**
- Billing granularity:
 - Per-second tracking
 - Rounded to per-minute for invoicing (configurable)

No per-seat pricing.

No feature-tier pricing.

3.2 Cost Components

Each call cost is derived from:

- Phone number region
- Direction (inbound / outbound)
- Duration
- AI processing overhead (optional markup)

Costs are **transparent and itemized**.

4. Usage Metering

4.1 Metered Events

- Call started
- Call connected
- Call ended
- AI processing time
- Human handoff (AI billing stops immediately)

Metering data must be:

- Immutable
- Time-stamped
- Attributed to:
 - Tenant
 - Campaign

- AI agent
 - Phone number
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4.2 Real-Time Usage Tracking

- Live minute consumption counter
- Tenant-visible usage dashboard
- Near-real-time balance updates (≤ 5 seconds lag)

No delayed surprises.

5. License Validity & Enforcement

5.1 License States

- Active
 - Expired
 - Suspended
 - Depleted (usage exhausted)
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5.2 Enforcement Rules (Confirmed)

- **Hard stop** when usage balance reaches zero:
 - Outbound calls blocked
 - Inbound calls receive system response
 - Campaigns auto-paused
- No grace usage
- No silent overdrafts

This rule is enforced at **API + execution layer**, not UI.

6. Invoicing

6.1 Invoice Generation

Invoices include:

- Billing period
- Total minutes consumed
- Cost breakdown by:
 - Number
 - Campaign
 - Call type
- Taxes (configurable)
- Payment status

Invoices are:

- Auto-generated
 - Immutable once issued
 - Downloadable (PDF)
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6.2 Invoice Frequency

- Monthly (default)
 - Custom periods supported (admin-configured)
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7. Payment Gateway Integration (Confirmed: External)

7.1 Supported Gateways

- Stripe
- Razorpay

Gateway selection is:

- Platform-level
 - Tenant-agnostic
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7.2 Payment Flow

- Invoice generated
- Payment link issued
- Payment confirmation webhook
- License renewed / usage credited

Failed payments trigger:

- Alerts
 - Grace period (optional)
 - Suspension (admin-configured)
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8. Prepaid vs Postpaid (Clarification Required)

Assumption (Default): Prepaid Usage Model

- Tenant prepays for minutes
- Balance deducted in real time
- Prevents credit risk

Postpaid can be introduced later but is **out of scope for v1 unless you override this assumption**.

9. Alerts & Notifications

Configurable alerts for:

- Low balance thresholds
- High usage spikes
- Invoice generated
- Payment failures

Delivery channels:

- In-app
 - Email
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10. Admin Controls

Super Admin can:

- Adjust balances (with audit reason)
- Override invoices
- Pause billing

- Issue credits
- View platform-wide revenue

All admin actions are:

- Logged
 - Auditable
 - Non-deletable
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11. Audit & Compliance

- Complete billing audit trail
- Reconciliation-ready reports
- Tax-compliant invoice numbering
- GDPR-safe storage of billing data

Billing data is retained per legal requirements.

12. Non-Functional Requirements (Module-Specific)

- Metering accuracy $\geq 99.99\%$
- Billing operations must not delay call execution
- Invoice generation ≤ 10 seconds
- Payment confirmation handling ≤ 5 seconds

Failure here is existential—this module must be rock-solid.

13. Out of Scope (Explicit)

- Discounts & promotions
 - Wallet transfers between tenants
 - Manual invoicing by customers
 - Multi-currency pricing (future)
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